



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, All UG Programmes are Accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

CIVIL ENGINEERING

(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

III/IV B.TECH

I-SEMESTER

(With effect from 2019-2020 Admitted Batch onwards)

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 HS 3101	Managerial Economics and Financial Accountancy	HS	3	3	--	--	25	75	100
B19 CE 3101	Structural Analysis-II	PC	3	3	--	--	25	75	100
B19 CE 3102	Design of Reinforced Concrete Structures	PC	3	3	--	--	25	75	100
B19 CE 3103	Soil Mechanics	PC	3	3	--	--	25	75	100
B19 CE 3104	Estimation, Specifications & Contracts	PC	3	3	--	--	25	75	100
# PE-I	Professional Elective-I	PE	3	3	--	--	25	75	100
B19 CE 3109	Soil Mechanics Lab	PC	1.5	--	--	3	20	30	50
B19 CE 3110	Environmental Engineering Lab	PC	1.5	--	--	3	20	30	50
B19 MC 3101	Employability Skills -I	MC	0	3	--	--	--	--	--
TOTAL			21	21	--	6	190	510	700

	Course Code	Course
#PE-I	B19 CE 3105	Repair, Rehabilitation and Retrofitting of Structures
	B19 CE 3106	Geo synthetics & Its Applications
	B19 CE 3107	Open Channel Flow
	B19 CE 3108	Advanced Water and Waste Water Engineering

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B19HS3101	HS	3	--	--	3	25	75	3 Hrs.
MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTANCY								
(Common to CE & IT)								
Course Objectives: Students are expected to learn								
1.	To Study Managerial Economics and Demand Analysis							
2.	To familiarize about the Concepts of Cost and Break-Even Analysis.							
3.	To understand the nature of markets and to know the Pricing Policies							
4.	To learn about accounting cycle and preparation of Financial Statements.							
5.	To know the concept of Capital and sources of raising and Depreciation							
Course Outcomes: At the end of the course the student will be able to								
S. No	Outcome							KL
1.	Equip oneself with the knowledge of estimating the Demand and demand elasticities for a product.							K2
2.	Have knowledge of Cost and its types and ability to calculate BEP							K3
3.	Understand the nature of different markets							K2
4.	Understand Pricing Practices prevailing in today's business world							K2
5.	Prepare Financial Statements and know how to calculate Profit & Loss for a firm							K3
6.	Know Types of capital and their sources and know how to calculate Depreciation							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Managerial Economics and demand Analysis: Managerial Economics: Definition of Economics & Classification of Economics (Micro & Macro), Meaning, Nature, & Scope of Managerial Economics.Demand Analysis: Concept of Demand, Determinants of Demand, Demand schedule, Demand curve, Law of Demand and its exceptions. Elasticity of Demand, Types of Elasticity of Demand. Importance of demand forecasting and its Methods.							
UNIT-II (10 Hrs)	Cost Analysis: Importance of cost analysis, Types of Cost- Actual cost Vs Opportunity cost, Fixed cost Vs Variable cost, Explicit Vs Implicit cost, Historical cost Vs Replacement cost, Incremental cost Vs Sunk cost; Elements of costs – Material, Labour, Expenses; Methods of costing - Job costing, contract costing, Process costing, Batch costing, Unit costing, Service costing, Multiple costing. Break-even analysis: Determination of Breakeven point - Applications, Assumptions and Limitations of Break -even analysis (Theory only).							
UNIT-III (10 Hrs)	Introduction to Markets & Pricing Policies Market Structures: Salient Features of Perfect Competition, Monopoly, Monopolistic competition, Oligopoly and Duopoly. Pricing: Importance of pricing and its meaning;Methods of Pricing: Cost Based -Full cost, Mark-up, Marginal &Break-even; Demand Based - Penetrating, Skimming; Competition Based- Going rate, Sealed Bid, Discount; Internet Pricing - Flat-rate, Usage sensitive.							

UNIT-IV (8 Hrs)	Introduction to Financial Accounting: Importance of Accounting - Double Entry System of Accounting - Types of Accounts - Journal, Ledger, Trail Balance, Trading Account, Profit and Loss Account and Balance Sheet (outlines only).
UNIT-V (12 Hrs)	Capital & Depreciation: Types of Capital - Fixed capital & Working Capital, Components of Working Capital, Factors influencing Working capital. Methods of Raising Finance - Short term, medium term and Long term. Depreciation - Meaning, Importance and causes of depreciation; Methods of Depreciation- Straight line and diminishing balancing methods (Theory only)
Text Books:	
1.	A R Aryasri, Managerial Economics and Financial Analysis, TMH Pvt. Ltd, New Delhi
2.	Dr. N.Appa Rao, Dr.P. Vijayakumar:Managerial Economics and Financial Analysis', Cengage Publications, New Delhi
Reference Books:	
1.	Dr.B.Kuberudu& T.V. Ramana : Managerial Economics and Financial anaysis, Himalaya Publishing House
2.	Varshney R.L, K.L Maheswari, Managerial Economics, S. Chand & Company Ltd,
3.	Shashi K. Gupta & R.K. Sharma Management Accounting, Kalyani Publishers
4.	Maheswari S.N, An Introduction to Accountancy, Vikas Publishing House Pvt Ltd

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3101	PC	3	--	--	3	25	75	3 Hrs.
STRUCTURAL ANALYSIS - II								
(For CE)								
Course Objectives:								
1	Analyse Indeterminate Truss by Force method and Energy Method							
2	Analyse Indeterminate Rectangular Portal Frame by Force method.							
3	Analyse Indeterminate Rectangular Portal Frame by Displacement method.							
4	Analyse Determinate and Indeterminate Arches.							
5	Analyse Cable system and Determinate and Indeterminate Suspension Bridges.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Determine the axial forces in the statically indeterminate trusses by Force method and Energy Method.							K3
2	Determine the member end moments and shears due to applied loads and yielding of supports for Statically Indeterminate Rigid Frame byForce method.							K3
3	Determine the member end moments and shears due to applied loads and yielding of supports for Statically Indeterminate Rigid Frame byDisplacement method.							K3
4	Determining the horizontal thrust and the vertical reactions at the supports, the orthogonal components axial thrust and radial shear and the resultant force at any point for three hinged and two hinged arches.							K3
5	Determine the shape of the cable, horizontal component of the axial tension in the cable and length of the cable. To determine the shear force and bending moment for three hinged and two hinged stiffening girder.							K3
SYLLABUS								
UNIT-I (08 Hrs)	Analysis of statically indeterminate trusses (having not more than 7 members and 3 supports) containing (a) external redundant supports (b) internal redundant members using (i) method of consistent deformation of unit load method (ii) Castigliano'stheorem – II.							
UNIT-II (06 Hrs)	Analysis of statically indeterminate frames (single storey, single bay portal frames only) and Induced Reactions on Statically Indeterminate Rigid Frame due to Yielding of Supports using (i) Method of consistent deformation (ii) Theorem of least work							

UNIT-III (10Hrs)	Analysis of statically indeterminate Rigid frames (single storey, single bay portal frames only) and Rigid frames due to Yielding of supports using (i) Slope-deflection method (ii) Moment distribution method (iii) Kani's method
UNIT-IV (8Hrs)	Arches: Introduction, Geometrical properties, Basic Mechanics, Arch action, Analysis of Arches for Normal thrust, radial shear and bending moment in three hinged and two hinged parabolic and segmental arches. Effects of yielding of supports, rib-shortening and temperature changes. Moving Loads and Influence Lines for Three-hinged and Two-hinged Arches.
UNIT-V (8 Hrs)	Cables and Suspension Bridges: Introduction, Properties of Suspended Cable, Analysis of Cable with concentrated loads and Uniformly distributed loads, Anchor Cable. Analysis of Suspension Bridges-Three-hinged Stiffening Girder- Two-hinged Stiffening Girder- Unstiffened Cables- Temperature Stresses in Three-hinged Stiffening Girder- Two-hinged Stiffening Gird; Influence Line Diagrams of Three-hinged Stiffening Girder- Two-hinged Stiffening Gird.
Text Books:	
1	Statically indeterminate structures – C.K. Wang, Mc Graw Hill Education PVT.LTD
2	Basic Structural Analysis, C.S. Reddy, Mc Graw Hill Education(India) PVT.LTD
3	Structural Analysis- T.S. Thandavamoorthy, Oxford University Press, New Delhi
Reference Books:	
1	Mechanics of structures Vol. II- S.B.Junnarkar and Dr.H.J.shah, Charotar Publishing House.
2	Structural analysis – Devdas Menon, Narosa Publishing House PVT.LTD
3	Theory of Structures Volume II, S.P Gupta, G.S Pandit, R. Gupta – Tata McGraw-Hill Publishing Company Limited, New Delhi

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3102	PC	3	--	--	3	25	75	3 Hrs.
DESIGN OF REINFORCED CONCRETE STRUCTURES								
(For CE)								
Course Objectives:								
1.	To introduce the different types of philosophies related to design of basic structural elements such as slab, beam, column and footing which form part of any structural system with reference to Indian standard code of practice.							
Course Outcomes (CO's): After the completion of the course, student will be able to								
S.No	Outcome							Knowledge Level
1.	Understand the various design methodologies for the design of RC elements. Analyse and design the flexural members.							K3
2.	Design the reinforced concrete beams subjected to shear only and also combined action of shear and torsion.							K3
3.	Distinguish between the behaviour of one way and two way actions in slab and familiarize to design of two way slabs whose corners restrained and not restrained from lifting up							K3
4.	Design compression members.							K3
5.	Design stair case and footing.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Loading standards as per IS 875, Grades of steel and cement, Introduction to basic design concepts like Working Stress Method (W.S.M), Ultimate Load Method (U.L.M) and Limit State Method (L.S.M.). Limit State of Collapse: Introduction, Characteristic load and strengths, Design values, Partial safety factors, Loads and materials, Stress-strain characteristics of concrete and steel. Limit State of Collapse in Flexure: Assumptions, Limiting depth of neutral axis. Concrete stress block in compression. Under reinforced, Balanced and over reinforced sections. Analysis of singly Reinforced rectangular sections, analysis of singly reinforced flanged section, analysis of doubly reinforced rectangular sections. Code requirement for design of flexural reinforcement are effective span, concrete cover, spacing of reinforcing bars, minimum and maximum areas of flexural reinforcement, requirements for deflection control, general guide lines for choosing beam size. Design of singly and doubly reinforced rectangular sections. Estimation of Effective flange width, Design of flanged beams (T-Beams).							
UNIT-II (10Hrs)	Limit State of Collapse in Shear: Shear stresses in homogeneous rectangular beam, modes of cracking, shear transfer mechanisms, shear span/depth ratio, and shear failure modes. Calculation of nominal shear stress, critical sections for shear design, Design shear strength							

	of concrete in beams with and without shear reinforcement. Types of shear reinforcement, Factors contributing to ultimate shear resistance, limiting ultimate shear resistance .Shear resistance of web reinforcement (Truss analogy), minimum shear reinforcement. Design of shear reinforcement in beams as per IS Code 456. Enhanced shear near supports. Steel detailing. Limit State of Collapse in Torsion: Need for torsion reinforcement, Reinforcement for torsion in RC beams, design strength in torsion combined with flexure and IS code provisions for design of longitudinal reinforcement, design strength in torsion combined with shear and IS code provisions for design of transverse Reinforcement, distribution of Torsion reinforcement. Design of rectangular section for combined bending shear and torsion. Detailing of torsion reinforcement. Limit State of Collapse in Bond: Concept of bond, Code requirement for bond, flexural bond, anchorage bond, development length. Bends, Hooks and Mechanical anchorages. Splicing of reinforcement, Lap Splices, Welded Splices and Mechanical Connections.
UNIT-III (10 Hrs)	Design of one way slabs: Behavior of one-way slabs, general considerations for slabs, minimum flexural reinforcement in slabs, deflection control by limiting Span/Depth ratio. Design of simply supported one way slabs and design of continuous one way slabs using moment coefficient, detailing of reinforcement in one-way slabs. Design of two way slab: Behavior of two-way slabs, design of wall supported two-way slabs, slab thickness based on deflection control criterion, Uniformly loaded and simply supported rectangular slabs (Rankine-Grash off theory), Uniformly loaded restrained rectangular slabs using IS 456 provisions. Detailing of flexural reinforcement and torsional reinforcement. Shear force in uniformly loaded two-way slabs.
UNIT-IV (10 Hrs)	Limit State of Collapse in Compression: Classification of columns based on type of reinforcement, type of loading and slenderness ratios. Estimation of effective length of a column-definition of effective length- unsupported length. Effective length ratios for idealized boundary conditions- Code recommendations for idealized boundary conditions. Code requirements on slenderness limits, minimum eccentricities and reinforcement. Design of short column under axial compression-condition of axial loading- behavior under ultimate loads-Tied columns-Spiral columns. Design strength of axially loaded short columns. Design of Short columns subjected to combined axial load and uniaxial moments by using design hand book SP:16. Design of Short columns subjected to combined axial load and biaxial moments by using design handbook SP:16.

UNIT-V (10 Hrs)	Design of Staircases: Types of staircases Geometrical Configurations, Structural Classification, stair slab spanning transversely, stair slab spanning longitudinally Loads and Load effects on stair Slabs, Design of stair slab spanning longitudinally. Design of Footings: Types of footing-Isolated footings, Soil pressures under isolated footings-Allowable soil pressure, Distribution of base pressure- Concentrically loaded footings, Instability problems-Overturning and sliding. General design considerations and Code requirements-Factored soil pressure at ultimate limit state, general design considerations, Thickness of footing base slab, Design for Shear, design for Flexure, Transfer of force at Column Base. Design of isolated square and Rectangular footings concentrically loaded.
Text Books:	
1.	Reinforced concrete design by S.Unnikrishna Pillai & Devdas Menon, Tata Mc. Graw Hill, New Delhi.
2.	Limit State Design by B.C.Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi publications Pvt. Ltd., New Delhi.
3.	Reinforced concrete Limit state design by Ashok K. Jain, Nem Chand & Bros, Roorkee.
4.	Limit state design of reinforced concrete – P.C.Varghese, Printice Hall of India, New Delhi.
5.	Reinforced Concrete Vol.I (Elementary Reinforced concrete) by Dr.H.J.Shah, Charotar publication house Pvt.Ltd, ANAND, Gujarat.
Reference Books:	
1.	Fundamentals of Reinforced concrete design by M.L.Gambhir, Printice Hall of India Private Ltd. ,New Delhi.
2.	Reinforced concrete structural elements—behavior, Analysis and design by P.Purushotham, Tata Mc.Graw-Hill, 1994.
3.	Design of concrete structures—Arthur H.Nilson, David Darwin, and Charles W.Dolar, Tata Mc.Graw-Hill, 3rd Edition, 2005.
4.	Reinforced concrete structures, Vol.1, by B.C.Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi, publications Pvt.Ltd., New Delhi
5.	Reinforced concrete structures— I.C.Syal & A.K.Goel, S.Chand Publishers
6.	Reinforced concrete design by N.Krishna Raju and R.N.Pranesh, New age International Publishers, New Delhi
7.	Design of Reinforced Concrete Structures by N. Subramanian, Oxford University Press, YMCA Library Building, 1 Jail Singh Road, New Delhi, India.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3103	PC	3	--	--	3	25	75	3 Hrs.
SOIL MECHANICS								
(For CE)								
Course Objectives: The objectives of this course are to make the student aware of								
1	Fundamental relationships between different parameters of a soil mass.							
2	Different types of soils and identify their properties.							
3	Processes of compaction and consolidation and apply them to field problems.							
4	Stress distribution, settlement, and strength of different soils in different conditions.							
5	Importance of Soil Mechanics in solving he engineering problems.							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1	Know the fundamental relationships between different parameters of soil mass and classify different types of soils along with identifying their properties							K3
2	Estimate Effective stresses and permeability of soils							K3
3	Estimate stress distribution in soil for different Load conditions							K3
4	Appreciate the processes of compaction and consolidation and apply them to field problems							K3
5	Identify shear strength parameters for different conditions							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Historical development, Soil Formation, Minerals in clays and sand, Soil Structure, Physical properties of Soil: Void ratio, Porosity, Degree of Saturation, Water content, Specific Gravity, weight–volume Relationships, Relative density, Consistency limits: Determination and consistency indices, Activity, Sensitivity and Thixotropy. Mechanical analysis and Soil Classification: Sieve analysis, Stokes law and hydrometer analysis. Unifiedsoilclassification,IndianStandardSoilClassificationSystems,Field Identification of Soils							
UNIT-II (8Hrs)	Soil Hydraulics: Types of soil water, capillary rise and surface tension, Darcy’s law and its limitations, constant head and variable head permeability tests, Factors effecting Coefficient of permeability, permeability of stratified soils. Total, neutral and effective stresses, Effective Stress Principle, Upward flow conditions, quick sand conditions and critical hydraulic gradient							

UNIT-III (8Hrs)	Stress Distribution in Soils: Bousinesq's theory for determination of vertical stress, assumptions and validity, rectangular and circular loaded areas, Pressure Bulb and Influence diagrams, westergaard's theory, Newmark's influence chart-construction and use, 2:1 approximate method, contact pressure distribution beneath footings.
UNIT-IV (12Hrs)	Compaction: Mechanism of compaction, Factors effecting compaction: water content, compaction effort, Type of soil. I.S Light and I.S Heavy compaction tests, Effect of compaction on soil Properties, Field compaction: compaction Equipment and Evaluation of field Compaction. Consolidation: Basic Definitions: compression index, coefficient of compressibility and coefficient of volume decrease. Terzaghi's one dimensional consolidation theory-assumption, Derivation of differential equation and Solution, Oedometer Test, Determination of coefficient of consolidation by time fitting methods, initial compression, primary compression and secondary compression, determination of pre consolidation pressure. Normally consolidated, over consolidated and under consolidated clays.
UNIT-V (12 Hrs)	Shear Strength of Soils: Stress at a point, Mohr circle of stress, Mohr coulomb failure theory, shear parameters, laboratory shear tests – shear box, tri axial and unconfined compression tests, laboratory and field vane shear tests, Sensitivity of clays, Types of shear tests on drainage conditions, shear strength of sands, critical void ratio and dilatancy, Factors affecting shear strength of clays and sands, Total stress analysis and Effective stress analysis.
Text Books:	
1	Soil Mechanics and Foundation Engineering by K.R. Arora.
2	Basic and Applied Soil Mechanics by Gopal Rajanand A.S.R.Rao.
3	Geotechnical Engineering by P .Purushothama Raj
Reference Books:	
1	Principles of Geotechnical Engineering– BrajaM.Das

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3104	PC	3	--	--	3	25	75	3 Hrs.
ESTIMATION, SPECIFICATIONS & CONTRACTS								
(For CE)								
Course Objectives:								
1.	To provide the student with the ability to estimate the quantities of item of works involved in buildings, water supply, road works and irrigation works							
2.	To Equip the student with the ability to do rate analysis							
3.	To develop the valuation of properties and preparation of reports for estimation of various items of work							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	List out various components, estimations and units of measurement for different works							K2
2	Apply the method of building estimate to find out the quantities of various items of work							K3
3	Determine the rate per unit of various items of work and their specifications							K3
4	Interpret the estimation of various roads and related items							K3
5	Select various methods to find out the valuation of a property & contracts							K3
SYLLABUS								
UNIT-I (8Hrs)	Introduction: Definition-Importance of estimation and costing, Standard units, Units of measurement of different items of work. ,Different types of estimates, Data required for preparation of estimate, Different technical terms in estimation- Contingencies, Work charged Establishments, measurement book, schedule of rates and related terms in the estimate, different types of approvals.							
UNIT-II (12Hrs)	Detailed estimate of buildings: Different items of work in building; Long wall and short wall method of building estimate, Centre line method of building estimate. Estimation of an RCC framed structure; Bar bending schedule- Beam and slab.							
UNIT-III (12Hrs)	Specifications: Meaning, purpose, types of specifications, general specification, detailed specifications of different items of buildings and other structures – Rate analysis –Data sheet for materials and various items of work in buildings and other structures, schedule of rates.							

UNIT-IV (8Hrs)	Estimate of earth work in roads; different formulae for calculations, Estimate of metalled road
UNIT-V (8 Hrs)	Valuation of buildings& Contracts: Purpose, different methods of building valuation; different terms used in valuation and their meaning,Types of contracts,Contract Documents,Conditions of contract.
Text Books:	
1.	Estimating and Costing in Civil Engineering by B.N. dutta.
2.	Estimation, Costing, Specifications and Valuation in civil Engineering by M.Chakraborti.
Reference Books:	
1.	Textbook of estimating and costing by G.S. Birdie.
2.	Textbook on Estimating, Costing and Accounts by D.D. Kohli and R.C. Kohli.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3105	PE	3	--	--	3	25	75	3 Hrs.
REPAIR, REHABILITATION AND RETROFITTING OF STRUCTURES								
(For CE)								
Course Objectives:								
1	To describe causes of distress in concrete structures and plan repair strategies.							
2	To explain issues on serviceability and durability of concrete							
3	To throw light on various repair materials and their characteristics							
4	To demonstrate repair techniques and protection measures							
5	To illustrate suitable retrofitting schemes.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Describe the reasons for deterioration in the concrete structures							K3
2	Evaluate the damage of concrete structures using various techniques like destructive and non-destructive tests							K3
3	Explain various parameters influencing the serviceability and durability of structures							K3
4	Understand the suitability of certain materials for a specific type of repair							K3
5	Recognize suitable techniques for repair and retrofitting.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Definition of Repair, Retrofitting, strengthening and rehabilitation. Types of failure/ damages in concrete structures. Causes of deterioration of concrete structures. Cracking-Types, causes and characteristics. Cracking in masonry walls and RCC structures.							
UNIT-II (10Hrs)	Damage Assessment: Purpose of assessment, Rapid assessment, Investigation of damage, Evaluation of surface and structural cracks, Damage assessment procedure, destructive, non-destructive, and semi destructive testing systems							
UNIT-III (10Hrs)	Influence on Serviceability and Durability of Concrete: Strength, Durability and Thermal properties– Effects due to climate, temperature, Corrosion - Effects of cover thickness and cracking. Methods of corrosion protection, corrosion inhibitors, corrosion resistant steels, coatings, and cathodic protection.							
UNIT-IV (10 Hrs)	Materials for Repair: Artificial fibre reinforced polymer like CFRP, GFRP, AFRP. Adhesive like Epoxy Resin, Special concretes and mortars, sulphur infiltrated concrete, polymer concrete, Expansive cement, Ferro cement, concrete chemicals, special elements for accelerated strength gain.							

UNIT-V (10 Hrs)	Techniques for Repair: Rust eliminators and polymers coating for rebars during repair, foamed concrete, mortar and drypack, vacuum concrete, Guniting and Shotcrete, Epoxy injection, Mortar repair for cracks, shoring and underpinning. Techniques for Retrofitting: Retrofitting of structural members i.e., column and beams by Jacketing technique, Externally bonding (ERB) technique, near surface mounted (NSM) technique, External post-tensioning, Section enlargement and guidelines for seismic rehabilitation of existing building.
Text Books:	
1	Poonam I. Modi, Chirag N. Patel, "Repair and Rehabilitation of Concrete Structures", PHI Learning private limited, Delhi.
2	J Bhattacharjee, "Concrete Structures: Repair, Rehabilitation and Retrofitting", CBS Publishers and Distributors Pvt. Ltd, New Delhi.
Reference Books:	
1	R T. Allen and S.C. Edwards, "Repair of concrete Structures", Blakie and sons, UK.
2	Santhakumar, A. R. "Training Course notes on damage assessment and Repair in Structures"
3	Raikar, R. N. "Learning from failures – deficiencies in Design, construction and service" R&D centre (SDCPL), Raikar Bhavan, Bombay.
4	D Campbell- Allen and Harold Roper, "Concrete Structures: Materials, Maintenance and Repair", Longman Scientific and Technical, U.K.
5	F. K. Garas, J. L. Clarke, G.S.T. Armer, "Structural Assessment", Butterworths, UK.
6	A.R. Santhakumar, "Concrete chemicals – Theory and applications, Indian society for construction Engineering and Technology", Madras

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3106	PE	3	--	--	3	25	75	3 Hrs.
GEO SYNTHETICS AND ITS APPLICATIONS								
(For CE)								
Course Objectives:								
1.	To create an awareness in the field of Geo synthetics							
2.	To impart the knowledge on manufacturing, properties and functions of Geo synthetics							
3.	To familiarize with the use of Geo synthetics in various civil engineering applications.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1.	Identify the type of Geo synthetics and their relevance in geotechnical field							k3
2.	Select suitable material for manufacturing of various types of Geo synthetics							k3
3.	Utilize the properties of Geo synthetics effectively in designing sustainable solutions							k3
4.	Select the type of Geo synthetics based on their function for its effective utilization							k3
5.	Apply Geo synthetics in various civil engineering applications for safer and economical constructions.							k3
SYLLABUS								
UNIT-I (8 Hrs)	An Overview of Geo synthetic in Geotechnical Engineering: Historical development, Types of geo synthetics: Geo textiles, Geo grids, Geo nets, Geo membranes, Geo synthetic clay liners, Geo cells, Erosion control products, Geo composites, Geo-others; Recent use in India.							
UNIT-II (8 Hrs)	Manufacturing - Materials and Process: Raw materials: polyamide, polyester, polyethylene, polypropylene, poly vinyl chloride; Different type of geo synthetics based on manufacturing - woven, monofilament, multifilament, slit filament, non-woven; Different bonding process: Mechanically bonded, Chemically bonded, Thermally bonded							
UNIT-III (12Hrs)	Properties of Geo synthetics: Physical properties, Mechanical properties, Hydraulic properties, Durability.							

UNIT-IV (10Hrs)	Functions of Geo synthetics: Reinforcement, Separation, Filtration, Drainage, Barrier, Protection; Suitability of various types of Geo synthetics according to function.
UNIT-V (12 Hrs)	Application of Geo synthetics: Soil Reinforcement: Reinforced soil walls, reinforced embankments, Reinforced soil beds, Pavements; Hydraulic Applications: erosion control, filtration and drainage; Lining
Text Books:	
1.	Engineering with Geo synthetics by G.VenkatappaRao and G.V.SSuryanarayanaRaju – Tata McGraw Hill, New Delhi, 1990
2.	An Introduction to Soil Reinforcement and Geo synthetics by G. L. Sivakumar Babu, University Press, 2005.
3.	Geo synthetics – An Introduction by G.V. Rao, Sai Master Geo environmental Services Pvt. Ltd. Hyderabad, 2011
4.	Designing with geo synthetics by R. M. Koerner, Pearson Education Inc., 2005
Reference Books:	
1.	Earth Reinforcement and Soil Structures ,Jones, C.J.F.P, Butterworth, 1985.
2.	Handbook on Geo synthetics and their applications by Sanjay Kumar Shukla, Thomas Telford, 2002
3.	Construction and Geotechnical Engineering using Synthetic Fabrics by Robert M.Koerner and Josoph P. Welsh. John Willey and Sons, New York.
4.	https://nptel.ac.in/courses/105/106/105106052/
5.	https://nptel.ac.in/courses/105/101/105101143/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3107	PE	3	--	--	3	25	75	3 Hrs.
OPEN CHANNEL FLOW								
(For CE)								
Course Objectives:								
1	To know the principles of fluid mechanics							
2	The solution of problems encountered in both natural and constructed water systems.							
3	Use of model studies and computers in solving a host of problems in hydraulic Engineering							
Course Outcomes:								
S.No	Outcomes							Knowledge Level
1.	Acquire a knowledge of the principles of mechanics of open surface flow of fluids, and able to express these in terms of mathematics.							K2
2.	Analyze problems associated with flow of water in streams and canals.							K4
3.	Develop the open channel flow equations from the basic conservation equation.							K4
4.	Solve open channel flow problems through the selection of appropriate equations and apply the physical mechanism of hydraulic jump.							K3
5.	Analyze gradually varied flow and design channel with hydraulic structures.							K4
SYLLABUS								
UNIT-I (10Hrs)	INTRODUCTION: Classification of flow in channels- steady flow, unsteady flow, uniform and non uniform flow, laminar and turbulent flow, sub-critical , critical and super-critical flow. Discharge through open channels by chezys formula, empirical formulae for the value of chezys constant.							
UNIT-II (10Hrs)	ECONOMICAL CHANNEL SECTIONS: Most economical rectangular section, most economical trapezoidal section, best side slope for most economical trapezoidal section, flow through circular channels, most economical circular section, most economical triangular section, open channel section for constant velocity at all depths of flow.							
UNIT-III (10Hrs)	SPECIFIC ENERGY: Specific energy and specific energy curve, critical depth, critical velocity, minimum specific energy in terms of critical depth, critical flow, streaming flow or sub-critical flow or tranquil flow, super-critical flow or shooting flow or torrential flow, alternate depths, condition for maximum discharge for a given value of specific energy.							

UNIT-IV (10Hrs)	HYDRAULIC JUMP: Definition, types of jumps, expression for depth of hydraulic jump, expression for loss of energy due to hydraulic jump, expression for depth of hydraulic jump in terms of Froude number, length of hydraulic jump, location of hydraulic jump, applications of hydraulic jump, mean velocity of flow in channels, metering flumes, surges in open channels.
UNIT-V (8Hrs)	GRADUALLY VARIED FLOW: Definition, equation of gradually varied flow, back water curve and afflux, expression for length of back water curve, classification of channel bottom slopes, classification of surface profiles, characteristics of surface profiles, Integration of varied flow equation, Measurement of discharges in rivers.
Text Books:	
1.	Hydraulics and fluid machines- Dr.P.NModi and Dr.S.M Seth
2.	French, R. H., "Open Channel Hydraulics", Water Resource Publication, First Edition, 2007
3.	Fluid mechanics and hydraulic machines- Dr.R.KBansal
4.	RangaRaju, K.G., "Flow through open channels" – Tata-McGraw Hill Education, 1981.
Reference Books:	
1.	Srivastava R, "Flow through Open Channels", Oxford University Press, New Delhi, 2007.
2.	Sturm T.W., "Open Channel Hydraulics", Tata-McGraw Hill Education, 2nd Edition, New Delhi, 2009
3.	Subramanya, K., "Flow in open channels" – Tata-McGraw Hill Education, 3rd Edition, New Delhi, 2009

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3108	PE	3	--	--	3	25	75	3 Hrs.
ADVANCED WATER AND WASTE WATER ENGINEERING								
(For CE)								
Course Objectives:								
1	Explain the needs of advanced methods in water and waste water treatment							
2	Understand the differences between conventional and advanced treatment methods							
3	Explain the biological kinetics required in advanced treatment systems							
Course Outcomes:								
S.No	Outcome							Knowledge Level
CO1	Outline the physical and chemical treatment processes for water							K2
CO2	Explain the principles of sedimentation, filtration and chlorination							K2
CO3	Explain the biological growth kinetics for waste water treatment							K2
CO4	Illustrate the working of advanced aerobic reactors for waste water treatment							K2
CO5	Illustrate the working of advanced anaerobic reactors for waste water management							K2
SYLLABUS								
UNIT-I (x Hrs)	Physical and Chemical Treatment: Screening, Grit removal, Aeration and gas transfer, Application of Membrane Processes, Reverse Osmosis, Micro-filtration, Nano-filtration, Ultra filtration and Electrodialysis, Chemical precipitation, other solids removal operations, Control of odour, Control of volatile organic compounds.							
UNIT-II (x Hrs)	Theory of Sedimentation – flocculent particle settling – theory of coagulation and flocculation- zeta potential - Filtration - theory of granular media filtration – head loss cleaning of filter media - backwash hydraulics – Theory of chlorination							
UNIT-III (x Hrs)	Principles: Objectives of biological treatment significance - aerobic and anaerobic treatment kinetics of biological growth - factors affecting growth – attached, suspended and Hybrid growth systems. Determination of kinetic coefficients for organics removal – Biodegradability assessment – selection of process – reactors – batch & continuous type							
UNIT-IV (x Hrs)	Aerobic Treatment of Wastewater: Sequential Batch Reactors (SBR), Stabilization ponds, Hybrid reactors for the treatment of wastewater :– MBBR, MBR, Expanded / fluidized bed bio reactors							

UNIT-V (x Hrs)	Anaerobic Treatment of Waste Water: Sludge handling and treatment -Sludge digestion: theory and principles - Disposal of digested sludge, Anaerobic ponds, UASB reactors and various modifications in UASB process and anaerobic filters
Text Books:	
1	Wastewater Engineering, Treatment and Reuse. Metcalf & Eddy, Inc. Tata McGraw-Hill Publications
2	Water and wastewater technology by Hammer and Hammer, PHI Publications
Reference Books:	
1	Biological Processes Design for wastewaters, Benefield, L.D. and Randall C.W. Prentice-Hall, Inc
2	Wastewater treatment for Pollution Control by Arceivala, Tata McGraw Hill Publication

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3109	PC	--	--	3	1.5	20	30	3 Hrs.
SOIL MECHANICS LAB								
(For CE)								
Course Objectives: Student shall be able to								
1	Tests to identify the physical properties of soil.							
2	Tests to identify the types of soil.							
3	Tests to identify compaction characteristics and permeability of soil.							
4	Tests to identify in-situ characteristics of soil.							
5	Tests to determine shear strength of soil.							
6	Tests to determine relative density of soil.							
7	Tests to determine of CBR of soil.							
Course Outcomes: After the completion of the course student should be able to								
S.No	Outcome							Knowledge Level
1	Identify the physical properties of soil and classify various types of soil							K2
2	Determine the permeability of soil							K3
3	Determine compaction characteristics of soils and Estimate in-situ density of soil							K3
4	Determine the shear strength parameters of soils by various methods							K3
5	Estimate the California Bearing Ratio (CBR)of a soil							K3
6	Determine the relative density of a coarse-grained soil							K3
LIST OF EXPERIMENTS								
1	Atterberg’s limits							
2	Field density by Core Cutter method.							
3	Field density by Sand replacement method							
4	Grain size analysis (Sieveanalysis)							
5	Hydro meter analysis.							
6	Specific gravity by pycnometer/density bottle method.							
7	Permeability of soil– Constant head method.							
8	Permeability of soil–Variable head method							
9	IS light/heavy compaction.							
10	CBR test /plate bearing test							
11	Unconfined compression test							
12	Tri axial compression test							
13	Direct shear test							
14	Vane shear test							
15	Relative density							
*At least 12 experiments must be done								
Reference Books:								
1	Soil Mechanics and Foundation Engineering by K.R. Arora							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3110	PC	--	--	3	1.5	20	30	3 Hrs.
ENVIRONMENTAL ENGINEERING LAB								
(For CE)								
Course Objectives: Student shall be able to								
1	To test physical properties of water							
2	To test chemical properties of water							
3	To know about the water quality testing as a whole							
Course Outcomes: After the completion of the course student should be able to								
S.No	Outcome							Knowledge Level
1	Determine physical properties of water							K5
2	Determine hardness, acidity and alkalinity of water							K5
3	Estimate chloride, available chlorine, BOD and COD							K5
4	Estimate solids present in water sample							K5
LIST OF EXPERIMENTS								
1	Determine pH of water							
2	Determine Hardness of water							
3	Determine Acidity of water							
4	Determine Alkalinity of water							
5	Determine Chlorides in water sample							
6	Determine DO in water sample							
7	Determine BOD in water sample							
8	Determine Estimation of total solids, suspended solids, dissolved solids in water sample							
9	Jar test							
10	Electrical conductivity							
Reference Books:								
1	Environmental Engineering by S.K.Garg							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3101	MC	3	--	--	--	--	--	--
EMPLOYABILITY SKILLS I								
Common to All Branches								
Part-A: Verbal and Soft Skills-I								
Course Objectives:								
1.	To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.							
2.	To familiarize the learner with high frequency words as they would be used in their professional career.							
3.	To inculcate logical thinking in order to frame and use data as per the requirement							
4.	To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.							
5.	To familiarize students with soft skills and how it influences their professional growth.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.							K3
2	Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.							K3
3.	Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.							K3
4.	Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.							K3
5.	Apply soft skills in the work place and build better personal and professional relationships making informed decisions.							K3
SYLLABUS								
UNIT-I	Grammar: (VA) Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns- number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.							
UNIT-II	Vocabulary: (VA) Synonyms and synonym variants (with emphasis on high frequency words), antonyms and antonym variants (with emphasis on high frequency words), contextual meanings with regard							

	to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.
UNIT-III	Reasoning: (VA) Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).
UNIT-IV	Usage: (VA) Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.
UNIT-V	Soft Skills: Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis – Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.
Text Books:	
1.	Oxford Learners,,s Grammar – Finder by John Eastwood, Oxford Publication.
2.	R S Agarwal,,s books on objective English and verbal reasoning
3.	English Vocabulary in Use- Advanced , Cambridge University Press
4.	Collocations In Use, Cambridge University Press
5.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi
Reference Books:	
1.	English Grammar in Use by Raymond Murphy, CUP
2.	Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	The Art of Public Speaking by Dale Carnegie
5.	The Leader in You by Dale Carnegie
6.	Emotional Intelligence by Daniel Golman
7.	Stay Hungry Stay Foolish by Rashmi Bansal
8.	I have a Dream by Rashmi Bansal.

Part-B: Quantitative Aptitude-I		
Course Objectives:		
1.	To familiarize students with basic problems on numbers and ratios problems.	
2.	To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.	
3.	To enable the students to work efficiently on percentage values related to shares, profit and loss problems.	
4.	To inculcate logical thinking by exposing the students to reasoning related questions.	
5.	To inculcate logical thinking by exposing the students to reasoning related questions.	
Course Outcomes:		
S. No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating on number problems and various units of ratio concepts	K3
2.	Accurate solving problems on time and distance and units related solutions	K3
3.	The students will become adept in solving problems related to profit and loss, in specific, quantitative ability	K3
4.	The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry	K3
5.	The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies	K3
SYLLABUS		
UNIT-I	Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion	
UNIT-II	Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.	
UNIT-III	Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends. Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.	

UNIT-IV	Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.
UNIT-V	Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.
Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal
3.	Puzzles to puzzle you by shakunataladevi.
References:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for cat by arunsharma.
5.	Elementary and Higher algebra by HS Hall and SR knight.
Websites:	
1.	www.m4maths.com
2.	www.Indiabix.com
3.	www.800score.com
4.	Official GRE site
5.	Official GMAT site

CIVIL ENGINEERING
(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R19)
III/IV B.TECH
II-SEMESTER
(With effect from **2019-2020** Admitted Batch onwards)

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 CE 3201	Design of Steel Structures	PC	3	3	--	--	25	75	100
B19 CE 3202	Transportation Engineering-I	PC	3	3	--	--	25	75	100
B19 CE 3203	Foundation Engineering	PC	3	3	--	--	25	75	100
B19 CE 3204	Air pollution and Control	PC	3	3	--	--	25	75	100
#PE-II	Professional Elective-II	PE	3	3	--	--	25	75	100
#OE-I	Open Elective-I	OE	3	3	--	--	25	75	100
B19 CE 3209	Highway materials testing Lab	PC	1.5	--	--	3	20	30	50
B19 CE 3210	Socially Relevant Project	PR	1	--	--	2	20	30	50
B19 MC 3201	Employability Skills –II	MC	0	3	--	--	--	--	--
B19 MC 3202	Basic Coding	MC	0	--	--	3	--	--	--
TOTAL			20.5	26	--	8	190	510	700

	Course Code	Course
#PE-II	B19 CE 3205	Pre stressed Concrete
	B19 CE 3206	Ground Improvement Techniques
	B19 CE 3207	Environmental Impact Analysis
	B19 CE 3208	Hydraulic Structures
#OE-I	Student has to study one Open Elective offered by CSE or ECE or EEE or IT or ME or EM &H from the list enclosed.	

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3201	PC	3	--	--	3	25	75	3 Hrs.
DESIGN OF STEEL STRUCTURES								
(For CE)								
Course Objectives: Student should be able to								
1	Understand fundamental concepts of limit state design of steel structures and design of bolted connection as per IS:800-2007							
2	Understand welded connection, weld specification and design of welded joints as per IS:800-2007							
3	Understand the behavior of the steel structure under tension and design of tension members as per IS:800-2007							
4	Understand the behavior of the steel structure under compression and design of compression members as per IS:800-2007.							
5	Understand the behavior of the steel structure under flexure and design of flexural members as per IS:800-2007.							
Course Outcomes: Student should be able to								
S.No	Outcome							Knowledge Level
1	Determine the number of bolts, pitch, gauge and strength of the joint by bolted connections.							K4
2	Determine the size of weld, length of weld, and strength of the joint by welded connections.							K4
3	Select suitable section as a tension member and determine the number of bolts, strength of the tension member.							K4
4	Select suitable section as a compression member and determine the strength of the axially loaded compression members as built up compression column with lateral supporting system.							K4
5	Select suitable rolled steel section as a flexural member and determine the flexural and shear strength and check the safety of the beam.							K4
SYLLABUS								
Note: All the designs should be taught in the limit state design method as per IS 800-2007”.								
UNIT-I (08 Hrs)	Fundamental Concepts of limit state design of structures, Different types of rolled steel sections available to be used in steel structures. Stress – Strain relationship for mild steel. Bolted connections: Behavior of bolted joints, Design strength of ordinary black bolts, high strength friction grip bolts, and Simple connections.							
UNIT-II (08 Hrs)	Welded Connections: Introduction, welding processes, Advantages of welding, Types and prosperities of welds, Types of joints, weld specifications as per IS 800:2007 code provisions, Types of weld defects, Design of lap joints and butt joints subjected to axial load using fillet and butt welds.							

UNIT-III (08 Hrs)	Tension members: Types of tension members, slenderness ratio, displacement of tension members, behavior of tension members, modes of failure, factors affecting strength of tension members, design of tension members, Lug angles.
UNIT-IV (10 Hrs)	Compression members: Possible failure modes, behavior of compression members, Effective length, and radius of gyration and slenderness of compression members, Allowable stresses in compression, Design of axially loaded compression members, built up compression members with lateral supporting system such as Lacing and Battened.
UNIT-V (06 Hrs)	Beams: Beam types, section classifications, lateral stability of beams, Allowable stress in bending, Shear and Bearing stresses, Effective length of compression flange, laterally supported and unsupported beams.
Text Books:	
1.	Design of Steel structures by N. Subramanian, Oxford University Press.
2.	Limit State Design of steel structures by S.K.Duggal, McGraw Hill Education Private Ltd.
3.	Limit State Design of steel structures – Ram chandra and Virendra Gehlot, Scientific Publishers (India)
Reference Books:	
1.	Design of steel structures by K.S.Sai Ram, Pearson Education India.
2.	Design of steel structures by Limit State Method as per IS: 800-2007 – S.S.Bhavikatti, IK Intern

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3202	PC	3	--	--	3	25	75	3 Hrs.
TRANSPORTATION ENGINEERING - I								
(For CE)								
Course Objectives:								
1	To impart different concepts in the field of Highway Engineering.							
2	To acquire design principles of Highway Geometrics and Pavements.							
3	To learn various traffic management plans.							
4	To acquire knowledge about pavement materials and design of flexible and rigid pavements.							
5	To learn various highway construction and maintenance procedures.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Plan the alignment of highway network for the given area.							K2
2	Design the highway geometrical elements.							K3
3	Design intersections and prepare traffic management plans.							K3
4	Identify the suitability of pavement materials and design flexible & rigid pavements.							K3
5	Understand the principles of construction and maintenance of highways.							K3
SYLLABUS								
UNIT-I (8Hrs)	Highway Planning and Alignment: Highway development in India; Classification of Roads; Road Network Patterns; Necessity for Highway Planning; Different Road Development Plans – First, second, third road development plans, road development vision 2021, Rural Road Development Plan – Vision 2025; Planning Surveys; Highway Alignment- Factors affecting Alignment- Engineering Surveys – Drawings and Reports.							
UNIT-II (8Hrs)	Highway Geometric Design: Importance of Geometric Design- Design controls and Criteria Highway Cross Section Elements- Sight Distance Elements-Stopping sight Distance, Overtaking Sight Distance and Intermediate Sight Distance- Design of Horizontal Alignment-Design of Super elevation and Extra widening- Design of Transition Curves- Design of Vertical alignment Gradients- Vertical curves.							
UNIT-III (10Hrs)	Traffic Engineering: Basic Parameters of Traffic-Volume, Speed and Density- Traffic Volume Studies; Speed studies –spot speed and speed & delay studies; Parking Studies; Road Accidents - Causes and Preventive measures - Condition Diagram and Collision Diagrams; PCU Factors, Capacity of Highways – Factors Affecting; LOS Concepts; Road Traffic Signs; Road markings; Types of Intersections; At-Grade Intersections – Design of Plain, Flared, Rotary and Channelized Intersections; Design of Traffic Signals –Webster Method –IRC Method.							

UNIT-IV (8Hrs)	Highway Materials: Sub grade soil: classification –Group Index – Sub grade soil strength – California Bearing Ratio – Modulus of Sub grade Reaction. Stone aggregates: Desirable properties – Tests for Road Aggregates – Bituminous Materials: Types – Desirable properties – Tests on Bitumen – Bituminous paving mixes: Requirements – Marshall Method of Mix Design. Design Of Pavements: Types of pavements; Functions and requirements of different components of pavements; Design Factors Flexible Pavements: Design factors – Flexible Pavement Design Methods – CBR method – IRC method – Burmister method – Mechanistic method – IRC Method for Low volume Flexible pavements. Rigid Pavements: Design Considerations – wheel load stresses – Temperature stresses – Frictional stresses – Combination of stresses – Design of slabs – Design of Joints – IRC method – Rigid pavements for low volume roads – Continuously Reinforced Cement Concrete Pavements – Roller Compacted Concrete Pavements.
UNIT-V (6 Hrs)	Highway Construction and Maintenance: Types of Highway Construction – Earthwork; Construction of Earth Roads, Gravel Roads, Water Bound Macadam Roads, Bituminous Pavements and Construction of Cement Concrete Pavements. Pavement Failures, Maintenance of Highways, pavement evaluation, strengthening of existing pavements.
Text Books:	
1	Highway Engineering, Khanna, S.K., Justo, C.E.G and Veer aragavan, A, Revised 10th Edition, Nem Chand & Bros, 2017.
2	Traffic Engineering and Transportation Planning, Kadiyali L. R, Khanna Publishers, NewDelhi.
Reference Books:	
1	Principles of Highway Engineering, Kadiyali L. R, Khanna Publishers, NewDelhi
2	Principles of Transportation Engineering, ParthaChakroborthy and Animesh Das, PHI Learning Private Limited, Delhi
3	Highway Engineering, Paul H. Wright and Karen K Dixon, Wiley Student Edition, Wiley India (P) Ltd., New Delhi
4	Transportation Engineering - An Introduction, JotinKhisty C, Prentice Hall, Englewood Cliffs, New Jersey.
5	Traffic & Highway Engineering by Nicholas J. Garber, Lester A. Hoel, Fifth Edition, published in 2015,CENGAGE Learning, New Delhi.
6	Transportation Engineering and Planning, Papacostas C.S. and P.D. Prevedouros, Prentice Hall of India Pvt.Ltd; New Delhi.
7	Highway Engineering, Srinivasa Kumar R, Universities Press, Hyderabad
8	Practice and Design of Highway Engineering, Sharma S. K., Principles, S. Chand & Company Private Limited, New Delhi.
9	Highway and Traffic Engineering, Subhash C. Saxena, CBS Publishers, New Delhi.
10	Transportation Engineering Volume I by C. Venkatramaiah, Universities Press, New Delhi.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3203	PC	3	--	--	3	25	75	3 Hrs.
FOUNDATION ENGINEERING								
(For CE)								
Course Objectives: The objectives of this course are to make the student aware of								
1	Different techniques of soil exploration.							
2	Various methods for estimating bearing capacity of different types of foundations.							
3	Estimating Load capacity of single piles and groups of piles.							
4	Settlement analysis of foundations.							
5	Theoretical aspects of well foundations.							
6	Concepts of stability of finite and infinite slopes.							
7	Earth pressures on retaining walls by Rankines and Coulombs theories							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1	Plan a detailed soil exploration program							K2
2	Apply various methods for estimating bearing capacity of different types of foundations.							K3
3	Estimate load capacity of single piles and groups of piles and know the theory aspects of well foundations							K3
4	Determine the stability of finite and infinite slopes.							K3
5	Calculate earth pressures on retaining walls using Rankine's and Coulomb's theories							K3
SYLLABUS								
UNIT-I (6 Hrs)	Sub soil Exploration: Methods of sub soil exploration Direct ,semi direct and indirect methods, Soundings by Standard, Dynamic cone and static cone penetration tests, Types of Boring, Types of samples, Criteria for undisturbed samples, Transport and preservation of samples, Bore logs, planning of exploration program, report writing.							
UNIT-II (12Hrs)	Shallow Foundations: Factors effecting location of foundation and design considerations of shallow foundations. Bearing Capacity: Safe bearing capacity and allowable bearing pressure, General and local shear failures, Terzaghis bearing capacity equations its modifications for square, rectangular and circular foundations, Factors affecting bearing capacity of Soil, Effect of water table on bearing capacity, IS Code method for Bearing capacity of footings, Allowable bearing pressure based on N values, Field plate load tests, Settlement Analysis of shallow foundations.							

UNIT-III (12Hrs)	Pile Foundations: Types, Construction, load carrying capacity of single piles in sands and clays(α-method) Dynamic Formula, Static formula, Pile load tests, Load carrying capacity of pile groups, Settlement Analysis of pile foundations. Caissons: Types of caissons, pneumatic caissons, Different shapes of well foundations, Relative advantages and disadvantages, Different Components of well and their function, Grip length, problems in well sinking and remedial measures.
UNIT-IV (10Hrs)	Stability Analysis of Slopes: Infinite and Finite Slopes, Stability Analysis of Infinite Slopes, different factors of safety, Types of Slope Failures – Toe, slope and Base failure, Stability Analysis of Finite slopes – Swedish Circle method, Friction Circle method, Fellenius method for location of Critical Slip Circle, Taylors stability number.
UNIT-V (10Hrs)	Earth Pressure: Types of Earth pressure, Rankness Active and passive earth pressure, Smooth Vertical wall with horizontal and inclined backfills. Coloumb's wedge theory, Culmans active earth pressure.
Text Books:	
1	Basic and Applied Soil Mechanics by Gopal Ranjanand A.S.R.Rao
2	Foundation Analysis and Design –J.E.Bowles.
3	Soil Mechanics and Foundation Engineering–ByK.R.Arora
Reference Books:	
1	Principles of Foundation Engineering- BrajaM.Das
2	Foundation Engineering-P.C.Verghese

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3204	PE	3	--	--	3	25	75	3 Hrs.
AIR POLLUTION AND CONTROL								
(For CE)								
Course Objectives:								
1	Explain the sources, characteristics, effects of air pollution							
2	Understand the meteorological conditions, plume behavior							
3	Understand the various pollution control equipment's and their processes.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Explain the classification, sources and effects of air pollution							K2
2	Explain the different meteorological conditions that influence the dispersion of the pollutants							K2
3	Illustrate the plume behaviour for different atmospheric stability conditions							K2
4	Adapt various pollution control equipment's or methods to control the discharge of pollutants							K3
5	Measure the pollution levels by sampling and analysis							K4
SYLLABUS								
UNIT-I (10 Hrs)	Air Pollution and its definition – Factors influencing air pollution – Classification of pollutants particulates – Gases – sources of pollution – Air quality standards – effects – Location of industries.							
UNIT-II (10 Hrs)	Meteorology – Wind roses – lapse rates – mixing depth atmospheric dispersion – plume behaviour accumulation, estimation of pollutants – effective stack height.							
UNIT-III (10 Hrs)	Air pollution effects on human beings, animals, plants and materials – Air pollution Episodes in India and Abroad.							
UNIT-IV (10 Hrs)	Air Pollution Sampling and measurement Ambient air quality monitoring and stack monitoring, Collection of particulate and gaseous pollutants: Isokinetic sampling: Ambient air quality survey.							
UNIT-V (10 Hrs)	Control of air pollution – removal of pollutants – particulate and gaseous – Air pollution control equipments (units) such as setting chamber, cyclones, wet scrubbers/collectors, scrubbers, centrifugal scrubbers, spray towers, packed beds, electrostatic precipitators, after burners – absorption – adsorption – diffusion.							
Text Books:								
1	Air pollution Control Technology by C.S.Rao.							
2	Air Pollution Control by K.V.S.G. Murali Krishna							
Reference Books:								
1.	Elements of Air Pollution Control by Prof. T. ShivajiRao							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3205	PE	3	--	--	3	25	75	3 Hrs.
PRE-STRESSED CONCRETE								
(For CE)								
Course Objectives:								
1	Analysis and design of pre stressed concrete members and connections.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Understand the general mechanical behaviour of pre stressed concrete.							K2
2	Analyze and design pre stressed concrete flexural members.							K3
3	Analyze and design for vertical and horizontal shear in pre stressed concrete.							K3
4	Analyze transfer and development length as well as pre stress losses.							K3
5	Analyze and design for deflection and crack control of pre stressed concrete members.							K3
SYLLABUS								
UNIT-I (08 Hrs)	Introduction to Pre stressed Concrete: Introduction, Basic concepts of pre stressing, need for high strength steel and concrete, advantages of pre stressed concrete. Materials for pre stressed concrete, high strength concrete and high strength steel. Pre stressing systems (1) Fressinet System (2) Gifford Udall (3) Magnel Blaton System, Tensioning devices, anchoring devices. (d) Pre tensioning and Post tensioning							
UNIT-II (08 Hrs)	Losses of Pre stress: Pre stressing losses, Loss Due to Elastic Deformation of Concrete, Loss due to shrinkage of Concrete, Loss Due to Creep of Concrete, Loss Due to Relaxation of stress in Steel, Loss Due to Friction, Loss Due to Anchorage Slip curvature Loss Due to Curvature etc. I.S. code provisions. Total Losses allowed for Design							
UNIT-III (08Hrs)	Analysis of Pre stress and Bending stresses: Basic Assumptions, Analysis of Pre stress, Resultant Stresses at a Section, Pressure Line or Thrust Line, Concept of Load Balancing, Stresses in Tendons							
UNIT-IV (10Hrs)	Flexural and Shear Strength of Pre stressed Concrete: Types of Flexural Failure, Strain Compatibility Method, Simplified Code Procedures, Shear and Principal Stresses, Ultimate Shear Resistance of Pre stressed Concrete Members.							
UNIT-V (06 Hrs)	Transmission of Pre stressing in Pre tensioned members: Transmission of Pre stressing Force by Bond, Transmission Length, Bond Stresses, Transverse Tensile Stresses, End-Zone Reinforcement, Flexural Bond Stress, I.S. Code Provisions for Bond and Transmission Length Anchorage Zone Stresses in Post-Tensioned members: Introduction, Stress Distribution in End Block, Investigations on Anchorage zone Stresses, Anchorage Zone Reinforcement.							

Text Books:	
1	Pre stressed Concrete by N Krishna Raju, 5 th edition, Tata McGraw Hill Publications.
2	Prestressed Concrete by Ramamrutham, 5 th edition, Dhanpatrai Publications
3	Fundamentals of Pre-stressed Concrete by Sinha N.C. and Roy S.K, 3 rd edition, S.Chand & Company Limited.
Reference Books:	
1	Design of Prestressed Concrete Structures by T.Y. Lin and Ned. H. Burns 3 rd edition, John Wiley & Sons
2	Pre-stressed concrete structures by N.Rajagopalan, 2 nd edition, Alpha Science International Ltd
3	Prestressed Concrete by P. Dayaratnam.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3206	PE	3	--	--	3	25	75	3 Hrs.
GROUND IMPROVEMENT TECHNIQUES								
(For CE)								
Course Objectives: The objectives of this course are to make the student aware of								
1	Various techniques of in-situ ground modification.							
2	Concepts, purpose and effects of grouting.							
3	Knowledge of reinforcement to soils in the form of geo textiles and geo grids.							
4	Basic knowledge on various ground improvement techniques and their suitability for various types of soil conditions.							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1	Apply in-situdensification methods for improving cohesive and cohesion less Soil deposits							K3
2	Apply grouting technique For improving soils							K3
3	Understand the purpose of geo textile and geo grid							K2
4	Apply the concepts of reinforced soil to various structures							K3
5	Understand various soil stabilization techniques							K2
SYLLABUS								
UNIT-I (14Hrs)	In-situdensification Methods: Granular soils–Introduction of vibration at the ground surface, Impact at the ground surface, vibration at depth, impact at depth, field compaction control; Cohesive soils- introduction, pre loading or de watering, drain walls, sand drains, sand wicks ,geo drains /band drains, lime columns, Stone columns: construction practice, comparison with lime column, design principles, vibro floatation techniques and other techniques like dynamic replacement etc., forced vacuum pre consolidation, thermal methods.							
UNIT-II (8Hrs)	Grouting: Introduction, grout injections, suspension and solution grouts, grouting equipment and methods, applications.							
UNIT-III (10Hrs)	Geo synthetics: Geo textiles: Introduction, types of geo textiles; Functions and their application, tests for geo textile, Geo grids: Introduction, types, functions and applications, tests for geo grids.							
UNIT-IV (8Hrs)	Reinforced Soil: Principles, components of reinforced soil, functions, determination of angle of inter facial friction, factors effecting angle of inter facial friction ,application of reinforced soil technique.							
UNIT-V (10 Hrs)	Stabilization: Mechanical stabilization, Soil aggregate mixture, properties and proportioning techniques, soft aggregate stabilization. Cement stabilization, Mechanism, factors affecting and properties, use of additives, design of soil cement mixtures, construction techniques.							

	Lime and Bituminous Stabilization: types of admixtures, mechanism, factors affecting , construction methods.
Text Books:	
1	Construction and Geotechnical Methods in Foundation Engineering, Robert M. Koerner : McGraw Hill.
2	Purushothama Raj.P,“Ground Improvement Techniques”, 2nded., Laxmi Publications(p) Ltd., New Delhi, 1998.
3	Engineering with Geosynthetics by G.VenkatappaRao and G.V.S.Suryanarayana Raju–TataMcGraw Hill, New Delhi, 1990.
Reference Books:	
1	Fundamentals of Geosynthetic Engineering by Sanjay Kumar Shukla, Jian-Hua Yin, CR C Press.
2	NPTEL: https://nptel.ac.in/courses/105108075/

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3207	PE	3	--	--	3	25	75	3 Hrs.
ENVIRONMENTAL IMPACT ANALYSIS								
(For CE)								
Course Objectives:								
1	learn the guidelines to prepare EIA and EIS report							
2	acquire knowledge in the methodologies and various attributes							
3	analyze the impact of urbanization and industrialization on the environment							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Explain the elements of EIA and EIS							K2
2	explain the environmental attributes to be considered for the EIA study							K2
3	explain the methodologies for EIA and review the relief and rehabilitation works							K2
4	explain the prediction and assessment of impacts							K2
5	Explain the case studies of EIA							K2
SYLLABUS								
UNIT-I (10Hrs)	EIA and EIS concepts – Elements of EIA – Guidelines for the preparation of EIS – Governmental policies for environmental protection							
UNIT-II (10Hrs)	Environmental attributes – air, water, soil, noise, ecological, social, economical, cultural, human and aesthetic aspects – Environmental indices							
UNIT-III (10 Hrs)	Methodology for the identification of impacts – Criteria for the selection of methods – Description of methodologies- Adhoc, Checklist, Overlaying, Matrix and Network methods							
UNIT-IV (10Hrs)	Prediction and Assessment of Impacts on – air, water, soil, noise, ecological, social, economical, cultural, human environments and aesthetic aspects							
UNIT-V (10Hrs)	Review of Environmental Impact Statement – Cost benefit analysis – Measures for environmental impact mitigation and control - Case studies							
Text Books:								
1	Environmental Impact Analysis – Canter							
2	Environmental Impact analysis – Urban and Jain							
Reference Books:								
1	Environmental Impact Assessment Methodologies by Y. Anjaneyulu and Valli Manikkam. BSP Books PVT Ltd.							
2	Environmental Impact assessment by Anji Reddy. BSP Books PVT Ltd							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3208	PE	3	--	--	3	25	75	3 Hrs.
HYDRAULIC STRUCTURES								
(For CE)								
Course Objectives:								
1	To provide planning and designing engineering concepts							
2	Application to various structures in hydropower projects including turbines and other Structures.							
Course Outcomes:								
S.No	Outcomes:							Knowledge Level
1.	Understand Reservoir planning and analysis							K2
2.	Analyze gravity and earth dams.							K4
3.	Design spillways and energy dissipation works.							K4
4.	Design Diversion head works							K4
SYLLABUS								
UNIT-I (12Hrs)	Reservoir Planning: Types of reservoir – Investigations for reservoir planning, Selection of site for a reservoir, Zones of storage in a reservoir; Purpose of reservoir, Reservoir regulation, Reservoir yield, Mass curve and Demand curve, Determination of reservoir capacity, Yield from a reservoir of given capacity; Economics of Water resources Projects –Apportionment of total cost of a Multi Purpose project.							
UNIT-II (10Hrs)	Dams: Investigation surveys, selection of dam site, selection of types of dam, classification of Dams. Gravity Dams Forces acting on dam, combination of forces for design, design parameters, design of gravity dam, profiles of a dam, stability analysis, foundation treatment, Galleries in gravity dams.							
UNIT-III (10Hrs)	Earth and Rock fill Dams: Types, design criteria for earth dams, design consideration in seismic region, phreatic line, flow net, stability analysis, methods of analysis, slope protection, seepage, dam section to suit available materials and foundation, causes of failure of earth dams, safety measures.							
UNIT-IV (8Hrs)	Spillways: Components factors affecting type and design of spillway, types, energy dissipation below spillways, hydraulic jump type stilling basins,spillway crest gates.							
UNIT-V (10Hrs)	Weirs and Barrages: Design of impervious floor on pervious foundation. Blighs creepTheory, potential theory cut offs, vertical drop weir design, Khosla's method.							
Text Books:								
1.	Creager, W.P, Justin, J and Daud Hinds, “Engineering for Dams”, Vol. I-III, Wiley, N.Y, USA.							
2.	Satyanarayana Murthy, C, "Design of Minor Irrigation and Canal Structure",Wiley Eastern, 1990.							

3.	Irrigation and water power engineering – Dr.B.C.Punmia
Reference Books:	
1.	Sharma, R.K, "Text Book of Irrigation Engineering and Hydraulic Structures ", -Oxford & IBH, 1984.
2.	Sharma, S.K, "Design of Irrigation Structures", S. Chand & Co, 1988.
3.	Varshney, R, S, “Theory and Design of Irrigation Structures”, Nem Chand & Bros, 2009.

Code	Category	L	T	P	C	I.M	E.M	Exam
#OE-I	OE	3	--	--	3	25	75	3 Hrs.
OPEN ELECTIVE-I								
(For CE)								
Student has to study one Open Elective offered by CSE or ECE or EEE or IT or ME or EM&H from the list enclosed.								

Offered by	Course Code	Course Name	Offered to
COMPUTER SCIENCE & ENGINEERING	B19CSOE01	Data Structures	CE,ECE,EEE, &ME
	B19CSOE02	Java Programming	
	B19CSOE02	Web Technologies	
ELECTRONICS & COMMUNICATION ENGINEERING	B19ECOE01	Basic Electronics	CE,CSE,EEE, IT &ME
	B19ECOE02	Signals & Systems	
ELECTRICAL & ELECTRONICS ENGINEERING	B19EEOE01	Introduction To Electrical Systems	CE,CSE,ECE, IT & ME
	B19EEOE02	Electrical Estimation and Costing	
INFORMATION TECHNOLOGY	B19ITOE01	Data Structures & Algorithms	CE,ECE,EEE, &ME
	B19ITOE02	Database Design & Development	
	B19ITOE03	Java Programming	
MECHNAICAL ENGINEERING	B19MEOE01	Operations Research	CE,CSE,ECE, EEE & IT
	B19MEOE02	Operations Management	
	B19MEOE03	Total Quality Management	
ENGINEERING MATHEMATICS & HUMANITIES	B19BSOE01	Computational statistics with R	ALL BRANCHES
	B19BSOE02	Fuzzy sets and Fuzzy Logic	

Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3209	PC	--	--	3	1.5	20	30	3 Hrs.
HIGHWAY MATERIALS TESTING LAB								
(For CE)								
Course Objectives: Student shall be able to								
1	To develop the testing skills of the road aggregates.							
2	To impart the knowledge on the properties of bitumen.							
Course Outcomes: After the completion of the course student should be able to								
S.No	Outcome							Knowledge Level
1	Characterize the highway aggregates.							K5
2	Evaluate the quality of Bitumen							K5
LIST OF EXPERIMENTS								
1	Determine the strength of the aggregates by using the Aggregate Crushing Value Test.							
2	Determine the toughness of the aggregate by using the Aggregate Impact Value Test.							
3	Determine the flakiness index of the given aggregate sample.							
4	Determine the elongation index of the given aggregate sample.							
5	Determine the Attrition value of a given aggregate sample.							
6	Determine the Abrasion value of a given aggregate sample.							
7	Determine the Specific gravity & Water absorption values for a given aggregate sample.							
8	Determine the Penetration value of a given bitumen sample.							
9	Determine the Softening point value for a given bitumen sample.							
10	Determine the Flash & fire point of a given bituminous material sample.							
11	Determine the Ductility value of a given bituminous material sample.							
12	Testing on Bituminous Mixes: Bitumen Extraction Test, Marshal Stability Test (Demonstration)							
Reference Books:								
1	Lab manual in High way Engineering by Ajay.K.Duggal& Vijay .P.Puri, New Age publications, New Delhi.							
2	Lab Manual prepared by Department of Civil Engineering, S.R.K.R Engineering College.							

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19CE3210	PR	--	--	2	1	20	30	3Hrs.
SOCIALLY RELEVANT PROJECT								
(For CE)								
Course Objectives: The aim of Socially Relevant Project is to encourage students								
1.	To express their ideas, to solve real-world problems and to complete projects							
2.	Using human experience to gather ideas from a wide range of problems in society by observation or pooling information							
3.	Using scientific, social-scientific, humanistic, cultural reasoning to analyze global problems							
Course Outcomes: The student learns to								
S.No	Outcome							Knowledge Level
1.	Use scientific reasoning to gather, evaluate, and interpret ideas							K4
2.	Analyze and design solutions to solve the ideas							K5
3.	Use one or more creative tools to complete the projects							K5
SYLLABUS								
Student can choose any one of the given below / any other socially relevant problem and work on it to produce a project document.								
1.	Water Conservation Related Works							
2.	Swatch Bharath (Internal External)							
3.	Helping police							
4.	Traffic monitoring							
5.	Teaching Rural Kids (Sarva siksha Abhiyan)							
6.	Street light monitoring							
7.	Electricity Conservation							
8.	Solar panel utilization							
9.	E- policing & cyber solution							
10.	Pollution							
11.	Any social related							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3201	MC	3	--	--	--	--	--	--
EMPLOYABILITY SKILLS II								
Common to All Branches								
Part-A: Verbal and Soft Skills-II								
Course Objectives:								
1.	To expose the students to bettering sentence expressions and also forming equivalents.							
2.	To instill reading and analyzing techniques for better comprehension of written discourses							
3.	To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.							
4.	To inculcate conversational skills, nuances required when interacting in different situations							
5.	To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.							K3
2	Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences							K3
3.	Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).							K3
4.	Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.							K3
5.	Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.							K3
SYLLABUS								
UNIT-I	Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.							
UNIT-II	Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author,,s intention							

	of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.
UNIT-III	Punctuation, discourse markers, general Essay writing, writing Issues and Arguments (with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a Statement of purpose Letters of Recommendation, business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.
UNIT-IV	Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.
UNIT-V	Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.
Reading/ Listening material:	
1.	Guide to IELTS, Cambridge University Press.
2.	Barron,s GRE guide
3.	Newspapers like The Hindu, Times of India, Economic Times.
4.	Magazines like Frontline, Outlook and Business India.
5.	News channels NDTV, National News, CNN
Text Books:	
1.	Objective English and Verbal Reasoning by R S Agarwal
2.	Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP
3.	Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi.
Reference Books:	
1.	Oxford Guide to Effective Writing and Speaking by John Seely.
2.	Collins Cobuild English Grammar by Collins
3.	The Art of Public Speaking by Dale Carnegie
4.	The Leader in You by Dale Carnegie
5.	Emotional Intelligence by Daniel Golman
6.	Stay Hungry Stay Foolish by Rashmi Bansal.

Part-B: Quantitative Aptitude-II		
Course Objectives: The objective of introducing quantitative aptitude –II is:		
1.	To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages	
2.	To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.	
3.	To familiarize the students with finding out accurate date and time related problems	
4.	To enable the students solve the puzzles using logical thinking.	
5.	To expose the students to various problems based on geometry and mensuration.	
Course Outcomes:		
S. No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating different types of data interpretation problems.	K3
2.	The students will perform efficaciously on analytical and logical problems using various methods.	K3
3.	Students will find the angle measurements of clock problems with the knowledge of calendars and clock.	K3
4.	The students will skillfully solve the puzzle problems like arrangement of different positions.	K3
5.	The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.	K3
SYLLABUS		
UNIT-I	Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.	
UNIT-II	Puzzle test, blood Relations, permutations, Combinations and probability, Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.	
UNIT-III	Periods, Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.	
UNIT-IV	Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles	
UNIT-V	Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles.	

	Types of triangles-Problems on triangles, Types of quadrilaterals- Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.
Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal. 3. Puzzles to puzzle you by shakunataladevi
3.	Puzzles to puzzle you by shakunataladevi
4.	More puzzles by shakunataladevi
5.	Puzzles by George summers
Reference Books:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for CAT by arunsharma
5.	Elementary and Higher algebra by HS Hall and SR knight.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3202	MC	--	--	3	--	--	--	--
BASIC CODING								
(Common to CE and ME)								
Course Objectives:								
1.	To develop programming skills among the students							
2.	To familiarize the student with Control Structures, Loop Structures							
3.	To familiarize the student with Basic searching and sorting Methods							
4.	To familiarize the student with Functions, Recursions and Storage Classes							
5.	To familiarize the student with Structures and Unions							
6.	To familiarize the student with Operating System concepts.							
7.	To familiarize the student with Networking concepts.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Know about Control Structures, Loop Structures and branching in programming							K3
2	Know about various searching and sorting methods.							K3
3.	Know about Functions, Recursions and Storage Classes							K3
4.	Know about Structures and Unions.							K3
5.	Know different Operating System concepts. 6.							K3
6.	Differentiate OSI Model Vs. TCP/IP suite.							K3
SYLLABUS								
UNIT-I	Review of Programming constructs Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.							
UNIT-II	Introduction to Linear Data, strings and pointers Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.							
UNIT-III	Functions, Recursions and Storage Classes Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.							

	Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime- memory allocation, Named locations Vs pointed locations, Referencing a 2D-Matrix.
UNIT-IV	User-defined data types, Pre-processor Directives and standard storage Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, Struct Vs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming. Practice: Structure padding, user-defined data storage and retrieval programs
UNIT-V	Operating system principles and Database concepts: Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer.
Reference Books:	
1.	Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2.	The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd.
3.	Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), EnzoMarinari (Author), Giovanni Organtini, World Scientific
4.	Object Oriented Programming in C++: N. Barkakati, PHI.
5.	Object Oriented Programming through C++ by Robat Laphore.
6.	https://www.geeksforgeeks.org/ .
7.	https://www.tutorialspoint.com/